11); contributed to the theory of the evaporation of black holes and its implications for the origin of black-hole entropy (Chapter 12); with Geroch, gave the first argument suggesting that time machines might be destroyed whenever they try to form (Chapter 14).

Weber, Joseph (b. 1919). American experimental physicist; invented the world's first gravitational-wave detectors ("bar detectors") and co-invented interferometric detectors for gravitational waves (Chapter 10); universally regarded as the "father" of the field of gravitational-wave detection.

Wheeler, John Archibald (b. 1911). American theoretical physicist; mentor for American researchers on black holes and other aspects of general relativity (Chapters 7); with Harrison and Wakano, developed the equation of state for cold, dead matter and a complete catalog of cold, dead stars, thereby firming up evidence that when massive stars die they must form black holes (Chapter 5); with Niels Bohr, developed the theory of nuclear fission (Chapter 6); led a team that designed the first American hydrogen bombs (Chapter 6); argued in a battle with Oppenheimer that black holes cannot form, then retracted the argument and became the leading proponent of black holes (Chapter 6); coined the phrases "black hole" (Chapter 6) and "a black hole has no hair" (Chapter 7); argued that the "issue of the final state" of gravitationally imploding stars is a key to understanding the marriage between general relativity and quantum mechanics, and in this argument anticipated Hawking's discovery that black holes can evaporate (Chapters 6 and 13); developed foundations for the laws of quantum gravity and, most important, conceived and developed the concept of quantum foam, which we now suspect is the stuff of which singularities are made (Chapter 13); developed the concept of the Planck—Wheeler length and area (Chapters 12, 13, 14).

Zel'dovich, Yakov Borisovich (1914–1987). Soviet theoretical physicist and astrophysicist; mentor for Soviet astrophysicists (Chapter 7); developed the theory of nuclear chain reactions (Chapter 5); invented key ideas that underlie Soviet atomic and hydrogen bombs, and led a bomb design team (Chapter 6); with Doroshkevich and Novikov, developed early evidence that a black hole has no hair (Chapter 7); invented several methods for astronomical searches for black holes, one of which seems ultimately to have succeeded (Chapter 8); independently of Salpeter, proposed that supermassive black holes power quasars and radio galaxies